

The road to *launch*

Five milestones from a suspended backend to 50 waitlist users trading — each independently shippable, each with a stated proof, none reachable by skipping the one before it. Plus the harness that keeps every future session of vibecoding inside this plan.

Doc	Date	Gate
06 of 06 — Execution roadmap	2026-08-24	.claude/verify.sh — green in 38s, every turn

R0 → R4: coherent before bigger, deployed before decorated

R0 - ✓ COMPLETE 2026-08-24

Stop the bleeding

All shipped on **fix/r0-stop-the-bleeding** : repo re-architecture (ae7f094) · demand-gate completion (f0dc04a) · **stream gate: scans-OR-exposure, idle = zero streams** + socket fixes (2896402) · upstream-channel refcount (176c08c) · plan decks (f45f248).

Proof: verify.sh green · 705 backend tests, 0 failed · idle+flat opens zero market streams · image contains no destructive scripts

R1 · RESUME & TRUTH

Production breathes

Runbook v2 (doc 05: env-first, resume ONLY two services, schema stamp) · m1 line becomes default branch · Pulse-dataclass fix · engine klines→REST · two-step engine enable (binary verified in logs, then gate) · pulse_snapshots writer + global pulse · **pulled forward from later: analysis-prompt shrink (40k→≤5k), fail-closed daily request counter, Redis eviction hardening.**

Proof: /health/ready green ·
market:pulse:* live · bandwidth <90
MB/day incl. LLM egress · RSS <400
MB · a real session proposes on live
pulses

R2 • THE PRODUCT

The free tier exists

Ordering is load-bearing:
rehydration (was R2.4) ships before the focus lock. Slot model — spent on proposal delivery, slot-native refunds · focus lock on reconciled state · channel gate (intraday) · onboarding 3–5 (capital / risk / **risk budget**) · Google-first · hot-window banner + /api/pulse · monitor_events table → trade-focus UI.

Proof: two users, same chart,
different proposals · restart mid-
trade → monitor + brackets intact ·
capacity flap x5 mints no extra scan
time

R3 • TRUST

Hardened for strangers

Kill NEXT_PUBLIC_API_TOKEN ·
anon fail-closed · WS auth off the
URL · /metrics behind token · global
LLM request counter · session/exit
recaps · forgot-password + copy
fixes.

Proof: security re-audit finds items 6–9 closed · a dead OpenRouter day degrades honestly, never silently

R4 • LAUNCH

Waitlist onboarded

Landing page: honest free-beta framing (no purchasable \$19/\$79) · invite cohorts of 10/day · metrics wired (activation, slots used, approval rate, monitor saves) · hot-window nudge email.

Proof: 50 invited · ≥60% complete onboarding + first session · zero quota or tenancy incidents in week one

SEQUENCING RULES (WHY THIS ORDER IS LOAD-BEARING)

- **R0 before resume** — resuming HEAD re-suspends the workspace in days; autoDeploy makes resume = deploy.
- **R1 before R2** — the hot-window banner needs live pulses and a snapshot history; slot mechanics need a running backend to verify against.
- **R3 before R4** — strangers, not the founder, will hold sessions open and poke the API; fail-open modes must be gone first.
- Real-money execution is **not on this map** — it gates on six weeks of live calibration + the compliance floor (doc 02, quant verdict).

FOUNDER ACTIONS (NOBODY ELSE CAN DO THESE)

- Render dashboard: settle overage → resume **ONLY** cryptai-backend + cryptai-redis (klaro + SlackBot **stay suspended** — hours cap)
- Supabase: verified **ACTIVE_HEALTHY** — nothing to do, just don't let the frontend go dark for 7+ days
- OpenRouter: one-time \$10 credit, then verify the real daily cap empirically
- Rotate the Render API key (leaked to chat 2026-08-03)
- Back up **VAULT_ENC_KEY** to a password manager
- Confirm the product decisions in doc 02, slide 7

R0 + R1: from suspended to live-and-bounded

ID	TASK	TEAM	DEFINITION OF DONE	ANCHOR
R0.1	DONE Repo re-architecture — 98 dead files attic'd, image cleaned, pointers fixed	infra	verify.sh green after the move; destructive scripts out of Docker context	ae7f094
R0.2	DONE Stream gate — demand = scans OR open exposure ; scan/prices/idle profiles, 120s linger, REST depth snapshots	backend	Idle+flat = zero streams; a position keeps one kline_1m per held symbol; 14 tests	2896402
R0.3	DONE Demand-gate completion — keep-warm decoupled from the switch; cadence aligned; envs documented	backend	10/10 demand-gate tests; render.yaml documents the new envs	f0dc04a
R0.4	DONE Idle-socket staleness skip · ws-callback dedupe · upstream-channel refcount	backend	No churn at zero subs; last watcher out → upstream released; 5 tests	176c08c
R1.1	Resume per runbook v2 — env deltas FIRST, resume only cryptai-backend + cryptai-redis, schema check + alembic stamp	founder+infra	/health/ready green; exactly one web service deployed; bandwidth <90 MB/day + RSS <400 MB over 48h; evicted_keys = 0	doc 05 · slide 4
R1.2	Make the m1 line the repo default branch; retire v2's stale keepalive; delete the TA-Lib CI step	infra	Scheduled workflows run the fail-loud keepalive from the default branch; CI ~6 min faster	keepalive.yml · ci.yml:74-87
R1.3	Fix the Pulse-dataclass bug + fold pulse liveness into the capacity check	backend	A test feeds real Pulse objects through the pipeline (not dict fakes); dead plane ends sessions with refund, never drains clocks	session_worker.py:216-235
R1.4	Engine klines → REST-pollled closed bars; then enable engine + pulse gating (two-step, logs between)	backend+infra	market:pulse:* live with ≤5s refresh; engine bandwidth ≤2 MB/day; sessions refuse honestly when pulse is stale	ingest.rs · doc 05 slide 4
R1.5	pulse_snapshots writer + market:pulse:global (median/max aggregate) + forward-return backfill job	backend+quant	Every publish appends a row with full factor JSON + reference price; backfill fills fwd returns; the calibration clock starts	FR-SIGNAL-2/3
R1.6	Pulled forward: prompt shrink 40k→≤5k tokens (LLM egress ≈ 1 GB/mo) + fail-closed daily request counter at ~80% of the live-verified cap	backend+quant	Prompt bytes measured before/after; setup quality spot-checked on 10 cycles; counter trip → 503 new + refund mid-scan, ops log names the cap	analysis_agent.py:1316
R1.7	Redis eviction hardening: candle bus messages persist=False; engine:on re-asserted; startup asserts cadence ≤ session cycle	backend	Queue payloads stop accumulating in the 25MB LRU; synthetic eviction of engine:on self-heals in ≤1 min	message_bus.py:120

R2: the 2-slot free tier, end to end

ID	TASK	TEAM	DEFINITION OF DONE	ANCHOR
R2.1	Rehydration first (was R2.4): engines rebuilt on boot from Postgres — positions, SL/TP bracket legs OCO-linked, original leverage (trade rows carry all three), desk balance; monitors reattach	backend	Kill the daemon mid-trade → monitor reattaches, a post-restart stop cross actually fills , balance intact	<code>paper_trading_engine.py:784</code>
R2.2	Plan model v2 + slot metering v2 : slot spent on proposal delivery (bounded ~10-min scan auto-returns it; ≤3 scans/day); slot-native refunds (CAPACITY_LOST re-opens its slot with the <i>remaining</i> grant); transactional consumption + one-active-session unique index; usage derived, no stored counter; boundary UTC, rendered local; sweeper ends disconnected sessions past grant	backend	Capacity flap ×5 mints no extra scan time; double-tap costs one slot; disconnected session ends at grant; a held slot survives one reset	<code>plans.py:57</code> · <code>session_manager.py:316</code>
R2.3	Focus lock (after R2.1): start refused on reconciled Redis+engine positions with zombie auto-drop; async tiers get a held-at-cap unmetered pause state (skipping synthesis alone still burns clock)	backend	409 distinguishes "monitored" from "reattaching"; capped async session accrues zero seconds; <code>/api/positions/close</code> stays the escape hatch	<code>server.py:1397</code> · <code>session_pipeline.py:150</code>
R2.4	Channel tier-gate: intraday everywhere; scalp not in the product; swing behind the +6-week validation gate	backend	<code>start(channel)</code> validates against <code>plan.allowed_channels</code> ; UI says why a channel is locked	<code>preferences.py:29</code>
R2.5	Onboarding steps 3–5 (capital → risk cards → monthly risk budget , never a return expectation) + Google-first reorder; desk epochs (capital edits blocked while a position is open)	frontend	All answers persist; resume-at-step works; no UI surface suggests a return number	<code>onboarding/page.tsx</code> · <code>TradingPersona.tsx</code>
R2.6	Hot-window surface: GET <code>/api/pulse</code> (per-symbol + global + "typically active hours" from the session-liquidity clock) + Desk banner + pulse strip	backend+frontend	Banner truthful states: hot now / quiet / engine warming / unavailable; never a forward prediction (quant framing)	<code>doc 03 slide 7</code>
R2.7	Trade-focus pivot — a derived UI state (position + monitor status; the session ENDED at <code>trade_opened</code>). Prerequisite: monitor_events append-only table + events endpoint — the <code>monitor_events</code> endpoint	backend+frontend	Approve → executing → watching automatically; verdicts persist across restarts; exit recap reads the trade's event trail from Postgres	<code>monitor_worker.py:327</code>
R2.8	Wire or delete the orphaned strategy cache (prompt shrink)	backend	Cache has a caller or is attic'd — never tested-but-unwired	<code>llm_cache.py:236</code>

The harness: rules the repo now enforces, not hopes

GATES (MECHANICAL)

- **.claude/verify.sh runs on every stop** — typecheck, 99 frontend + 687 backend tests, import smoke, 38s. A red gate sends the work back; the gate is never weakened to pass.
- **Branch per unit of work** (feat/ fix/ chore/); never commit to the default branch; small commits that say why.
- npm run build before push (7s warm); cargo test for engine changes (cargo lives at /opt/homebrew/opt/rustup/bin — not on default PATH).
- Docs of record: `CLAUDE.md` → `docs/ARCHITECTURE.md` → `docs/plan-2026-08/`. A session that contradicts them updates them or stops.

LAWS (THIS REPO'S SCAR TISSUE)

- **The wiring law:** "tested" ≠ "wired". No feature is claimed done without a constructor traced from server.py or the daemon. This repo's signature failure, three audits running.
- **The cost law:** every stream, loop, and LLM call names its live consumer, or it doesn't ship. Bytes/day × 30 vs the 5 GB ledger, in the PR description.
- **The money law:** never widen a risk gate, cap, or stop to get green; `ENABLE_EXECUTION` and `LIVE_TRADING_CONFIRMED` stay fail-closed; every query filters `user_id`.
- **The honesty law:** no fabricated numbers in the UI; no directional claim from regime; no forward "hottest hour" until logged data earns it.

SESSION RITUAL (FOR EVERY FUTURE AI CODING SESSION)

- **Open with:** read `CLAUDE.md` → the current milestone in doc 06 → pick ONE task ID (e.g. R2.4). Out-of-plan work needs a written reason in the commit.
- **Mid:** establish what runs before changing it; prefer deleting to commenting; new files must appear in `pytest --collect-only`.
- **Close with:** verify.sh green · task's definition-of-done demonstrated (command output or screenshot, not narrative) · `CLAUDE.md` updated if reality moved.
- Quarantined tests: read the failure before blaming the test — some encode real regressions.

R3 + R4 IN ONE LINE EACH

R3 (trust): retire the static admin token · anonymous fail-closed · WS auth out of the URL · /metrics behind auth · recap cards · forgot-password (the LLM request counter moved up to R1.6). **R4 (launch):** honest free-beta landing (no purchasable pricing) · cohorts of 10/day · product metrics live (activation, slots, approval rate, monitor saves) · hot-window nudge · week-one incident watch.

The whole team, one page

TEAM	NEXT	THEN	THEN
Product	Confirm the founder decisions (doc 02 s7); lock free-tier copy: "up to 2 sessions — a slot is only spent when it delivers a trade"	Landing page → honest free-beta; ladder markets only validated axes	Define week-one metric review ritual against the doc-02 targets
Backend	R0 DONE → R1.3–R1.5: Pulse bug, engine enable path, snapshots writer	R1.6–R1.7: prompt shrink, request counter, Redis hardening	R2.1–R2.4: rehydration, slots v2, focus lock, channel gate
Frontend / UX	Google-first reorder + forgot-password + stale-copy fixes (small, unblocks trust)	R2.5 onboarding steps + R2.6 hot-window banner	R2.7 trade-focus pivot + monitor verdict log + recaps
Infra / DevOps	R1.1 resume runbook with the founder; verify bandwidth graph daily for a week	R1.2 default-branch flip + keepalive + CI cleanup	Ops rhythm (doc 05 s5) + Sentry free tier when quiet
Quant	R1.5 snapshots writer spec: factor JSON, reference price, backfill horizons	+6 weeks: re-run walk-forward on live data; hot-hours from logged seasonality	Proposal→outcome ledger review; go/no-go evidence memo for real-money gate
Founder	Dashboard actions (doc 06 s2): resume, restore, \$10 credit, key rotation, vault backup	Decide the 5 open product calls	Invite cohort #1 of 10 — and trade your own product daily

THE SINGLE SENTENCE TO KEEP

Nothing new gets built while something already-built stays unwired — deploy truth first, wire second, decorate last. That discipline is the entire difference between the last six months and the next six weeks.